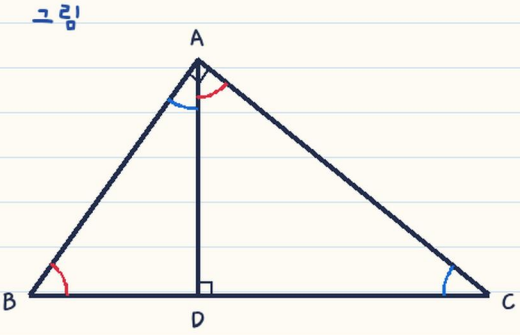
개념 설명

직각삼각형의 닮음과 삼각비

그림



$\triangle ABC \backsim \triangle DBA \backsim \triangle DAC$

(AA 닮음)

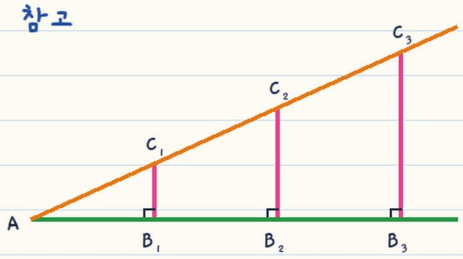
$\angle B = \angle DAC, \quad \angle C = \angle DAB$

$$\sin B = \frac{\overline{AC}}{\overline{BC}} = \frac{\overline{AD}}{\overline{AB}} = \frac{\overline{CD}}{\overline{AC}}$$

$$\cos B = \frac{\overline{AB}}{\overline{BC}} = \frac{\overline{BD}}{\overline{AB}} = \frac{\overline{AD}}{\overline{AC}}$$

$$\tan B = \frac{\overline{AC}}{\overline{AB}} = \frac{\overline{AD}}{\overline{BD}} = \frac{\overline{CD}}{\overline{AD}}$$

참고



$$\frac{\overline{B_1C_1}}{\overline{AC_1}} = \frac{\overline{B_2C_2}}{\overline{AC_2}} = \frac{\overline{B_3C_3}}{\overline{AC_3}} = \sin A$$

→ 각의 크기가 정해지면 삼각비의 값은 일정